Conference Paper Title*

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***Abstract*—Write this last, once your experiments are complete. It should contain five elements: 1. Context: Ball action spotting aims to identify and temporally localize ball-related events in broadcast soccer videos. 2. Problem: Accurate localization is challenging because events are brief and densely distributed. 3. Method: You investigate replacing the 3D CNN temporal encoder with a Transformer while retaining the 2D visual backbone. 4. Experiments: Describe the dataset, evaluation metric, and comparison. 5. Results and conclusion: Report your actual results and the main finding. Aim for approximately 150–200 words, depending on the conference requirements.**

***Index Terms*—Ball action spotting, soccer video analysis, temporal action localization, 3D convolutional neural networks, Transformers, video understanding.**

I. Introduction

Paragraph 1 — General problem Introduce video understanding and temporal action spotting.

Explain that action spotting involves identifying both the semantic category and the temporal location of events in long, untrimmed videos.

Then introduce soccer as an interesting application because broadcast videos contain many events, and understanding them requires temporal reasoning.

The improvements in deep learning and increase computing power permitted considerable progress in the field of video understanding. In particular sport video understanding field has become more important area for computer vision researcher [1] [2] [3] moving from easier task like object detection to more demanding challenges like temporal action spotting.

Temporal action spotting is a computer vision task that identifies the exact timestamp when a specific action or event begins or occurs in an untrimmed video, without needing to outline its full start-and-end duration.

Paragraph 2 — Ball action spotting

To advance research in soccer video understanding the SoccerNet Challenge was proposed at CVPR 2023. The challenge focused on 7 tasks divided into three categories: Broadcast video understanding, Field Understanding, and Player understanding. We will focus on the first edition of Ball action spotting challenge included in the first category.

A. Cioppa et al. [4] define ball action spotting challenge as localizing the exact time when two types of actions related to the soccer ball occur: pass and drive. This task pose a unique challenge due to the dynamic and complex nature of the game. During a match players move rapidly, often obscuring each other, making it difficult to accurately track their movements. In this challenge a precise timestamp is set for this actions:

- **pass event**: the timestamp correspond to the moment when the ball leaves a player's foot
- **drive event**: the timestamp corrrespond to the moment when a player control the ball

The dataset consist of 7 games with 11,041 annotated timestamps with 2 games not publicy available for the challenge set. The recordings of the games are untrimmed and provided by Footovision. Other details of the challenge can be found at SoccerNet Ball Action Spotting challenge.

Narrow the problem to ball-related events.

Explain: - What ball action spotting is. - The difference between general soccer action spotting and ball action spotting. - Why precise temporal localization matters. - Why ball actions can be difficult to distinguish. Mention the relevant SoccerNet challenge and dataset.

Paragraph 3 — Existing approaches and research gap Introduce the main approaches.

For example: - CNN-based approaches extract visual information from individual frames. - 3D convolutions can learn spatiotemporal features from neighboring frames. - Transformers use self-attention to model relationships between temporal representations. Explain that your work investigates whether self-attention provides a useful alternative to convolutional temporal modeling.

Be careful: don't claim that existing CNN approaches are inadequate. Your experiments should establish whether the alternative is beneficial. Paragraph 4 — Your approach Describe your idea briefly: - Start from the architecture of the 2023 winning solution. - Retain the shared 2D visual feature extractor. - Replace the 3D CNN temporal encoder with a Transformer encoder. - Compare the resulting architecture with the convolutional baseline. Contributions End the Introduction with a short contribution list.

The main contributions of this work are:

- We investigate a Transformer-based temporal encoder for ball action spotting, replacing the 3D convolutional component of an existing architecture.

- We describe an architecture that combines a shared 2D visual feature extractor with self-attention-based temporal feature aggregation.
- TODO: check this part We experimentally compare convolutional and Transformer-based temporal modeling in terms of spotting performance and computational efficiency.

In order to understand the salient actions of a broadcast soccer game, SoccerNet introduces the task of action spotting, which consists in finding all the actions occurring in the videos. Beyond soccer understanding, this task addresses the more general problem of retrieving moments with a specific semantic meaning in long untrimmed videos.
Introduce temporal action spotting in broadcast soccer. Explain that the model must identify both the action category and its timestamp in an untrimmed video, and that ball events can be brief and closely spaced. Introduce temporal 3D convolutions as a way to learn features across neighboring frames. Explain their local temporal processing and why you want to investigate self-attention as an alternative. Avoid claiming that convolution is inherently inadequate; your experiments should determine whether the alternative helps. Explain that your work investigates whether changing the temporal feature encoder, while preserving the rest of the pipeline, changes accuracy and efficiency on this particular task. The gap should be supported by related literature, not merely by the fact that you implemented a Transformer. List only contributions you actually make. For example:

- An implementation of a Transformer-based temporal encoder in an existing ball action spotting pipeline.
- A controlled comparison against a 3D CNN temporal encoder.
- An analysis of accuracy, temporal context, and computational cost.

II. Related Work

This section explains what has already been done and establishes where your work fits.

I recommend four subsections. 3.1 Soccer Action Spotting Introduce the broader problem of soccer action spotting.

Discuss: - SoccerNet and its datasets. - Temporal localization of soccer events. - The use of deep learning for action spotting. - Relevant approaches to temporal event detection. The objective is to give readers enough background to understand ball action spotting.

You don't need to describe every SoccerNet paper. Focus on the approaches most relevant to your architecture.

A. Ball Action Spotting

This is where you discuss the two references we examined.

Baikulov (2023)

Explain: - The slow-fusion 2D/3D architecture. - EfficientNetV2-B0. - The 3D inverted residual blocks. - Multi-stage training. - Transfer learning and fine-tuning with longer sequences. Explain why this is particularly relevant: it provides the architectural reference for your baseline.

Solution Key points:
Efficient model architecture for extracting information from video data Multi-stage training (transfer learning, fine-tuning with long sequences) Fast video loading for training (GPU based, no need for preprocessing with extracting images) Model The model architecture employed in this solution utilizes a slow fusion approach, incorporating 2D convolutions in the early stages and 3D convolutions in the later stages. This architectural choice made one of the most significant contributions to the overall metric outcome. It effectively improved the mAP@1 metric on both the test and challenge sets, resulting in an approximate increase of 15% (from 65% to 80%) compared to an approach utilizing 2D CNN early fusion.
model
The model consumes sequences of grayscale frames. Neighboring frames are stacking as channels for input to the 2D convolutional encoder. For example, if fifteen frames are stacked in sets of three, the outcome would be five input tensors, each consisting of three channels. The shared 2D encoder independently processes these input tensors, producing visual features. The following 3D encoder processes visual features, producing temporal features. Concated temporal features pass through global pooling to compress the spatial dimensions. Then, a linear classifier predicts the presence of actions in the middle frame of the original input sequence.
I choose the following model hyperparameters as a result of the experiments:

Stack threes from 15 grayscale 1280x736 frames skipping every second frame from the original 25 FPS video (equivalent to 15 neighboring frames at 12.5 FPS, about 1.16 seconds window) EfficientNetV2 B0 as 2D encoder 4 inverted residual 3D blocks as 3D encoder (ported from 2D EfficientNet version) GeM as global pooling Multilabel classification with positive labels applied in a 0.6 seconds window (15 frames at 25 FPS) around the action timestamp from annotation You can find more details in the model implementation and experiment configs.

Training
I made several stages of training to obtain 86.47% mAP@1 on the challenge set (87.03% on the test):

Basic training (config). The 2D encoder is initialized with ImageNet-pre-trained weights; other parts start from scratch. Training on Action Spotting Challenge dataset (config). Same weights initialization as in the first stage. Transfer learning (config). 2D and 3D encoders initialized with weights obtained in the second stage. Out-of-fold predictions from the first stage were used for data sampling (more details later). Fine-tuning with long sequences (config). 2D and 3D encoders initialized with weights obtained in the third stage. 2D encoder weights are frozen. Basic training In this challenge, I employed 7-fold cross-validation to tune the training pipeline more precisely. Each labeled game from the dataset is an individual fold.
In short, the resulting training pipeline hyperparameters are:

- Learning rate warmup for the first 6 epochs from 0 to

3e-4, cosine annealing last 30 epochs to 3e-6
- Batch size 4, one training epoch comprises 6000 samples
- Optimizer AdamW with weight decay 0.01
- Focal loss with gamma 1.2
- Model EMA with decay 0.999
- Initial weights for 2D encoder ImageNet pretrained
- Architecture hyperparameters listed above in the model part

Worth writing about sampling techniques during training, which significantly impacts its results. For basic training, a simple but well-working sampling algorithm was used. For each training sample, randomly take a video index from a uniform distribution. Then randomly choose a frame index with an equal chance to sample frames near to actions and remaining frames. Frame near to action if allocated in a 0.36 seconds window (9 frames at 25 FPS) around the action. I tried different ratios, but an equal chance to show empty and event frame worked best. I will introduce a more advanced sampling scheme in the section on transfer learning.

I applied the usual augmentations like horizontal flip, rotation, random resized crop, brightness, motion blur, etc. Of the interesting, there are only two temporal augmentations:

Simulating camera movement (change translation, scale, and angle over time) Randomly shake the frames in sequences (applied 40% chance to change the frame index on +- 1) The models from this training have 79.06% on CV (cross-validation) and 84.26% mAP@1 on the test set (the metric on test split was calculated by the out-of-fold predictions for two folds which include test games). I did not evaluate these models for the challenge set. **Training on Action Spotting Challenge dataset** I built a similar pipeline for videos and classes from Action Spotting Challenge to get good initial weights for the next experiment. Briefly, here are the changes from the previous:

- 377 games in training, 18 in validation split
- 15 classes (all cards-related class was merged because the model consumes grayscale frames)
- 4 warmup epochs and 20 training epochs, one epoch is 36000 samples
- Weight the frame sampling by the effective number of class samples
- Transfer learning

This training uses the results of the previous two. The second one gives excellent initial weights for 2D and 3D encoders. It provides a significant boost (2% mAP@1 on test and CV). That is understandable because the same models were trained on many games with similar input frames and some similar actions.

Basic training gives out-of-fold predictions that I use for sampling in the following way: **sampling**

Take the element-wise maximum between the sampling distribution (introduced above) and predictions, then normalize again to equal probability sums between empty and action frames. The intuition is that there are some hard negative examples in the dataset. Due to many negative samples, such hard examples are rarely sampled during training. We can make something like hard negative mining/sampling with the technique.

Other minor changes compared to basic training:

7 warmup epochs and 35 training epochs Focal loss with gamma 1.2 and alpha 0.4 Models achieve 81.04% on CV, 86.51% on the test, and 86.35% mAP@1 on the challenge set.

Fine-tuning with long sequences Before, I trained all models on relatively short clips (15 frames in 12.5 FPS, 1.16 seconds) to fits the VRAM of a single GPU and to decrease training time. Models obtained good 2D and 3D encoder weights in the previous experiment. So I tried to freeze 2D weights and fine-tune 3D weights on long sequences to provide more context.

Changes compared to transfer learning experiment:

33 frames in 12.5 FPS (2.6 seconds) LR warmup first 2 epochs from 0 to 1e-3, cosine annealing last 7 epochs to 5e-05 SGD with Nesterov momentum 0.9 Models scored 80.49% on CV, 87.04% on the test, and 86.47% mAP@1 on the challenge set. The score is lower on cross-validation, but it's my best submission on the test and the challenge set.

Prediction and postprocessing Models predict each possible sequence of frames from the videos. Additionally, I make test time augmentation with the horizontal flip. On the challenge set, I used the arithmetic mean of predictions from all fold models.

Postprocessing is very simple. I just used a combination of Gaussian filter and peak detection from SciPy with the following parameters: standard deviation for Gaussian kernel 3.0, peak detection minimal height 0.2, and minimal distance between neighboring peaks 15 frames.

Training and prediction accelerations I optimized the training pipeline to iterate experiments faster and to test more hypotheses.

- Custom multiprocessing video loader with simultaneous use VideoProcessingFramework (GPU decoding) and OpenCV (CPU decoding) workers to optimize hardware utilization
- FP16 with Automatic Mixed Precision
- torch.compile using TorchDynamo backend
- Augmentation on the GPU with kornia

These accelerations allow running epoch (train + val) of basic training in 7 minutes and 10 seconds on a single RTX 3090 Ti. It is impressive because one epoch is 6000 training and approximately 2600 validation examples, each of which is 15 frames in 1280x736 resolution. Also, using source videos without the preprocessing with extracting images allows using any video frame during training and saves disk space.

I applied caching strategy to speed up inference time using the architecture structure. If one saves the last visual features, it is enough to predict with the 2D encoder only one stack of frames when receiving a new one. The 2D encoder is the most time expensive part of the model. Predicting 3D features takes a short time. So this strategy dramatically boosts prediction speed by several times.

My solution is very inspired by the top solutions of the DFL - Bundesliga Data Shootout competition: Team Hydrogen (link) K_mat (link) Camaro (link) ohkawa3 (link)

B. Wang et al. (2023)

Authors summary

Here we talk about the solution with the paper.

Our method, Boosted Model Ensembling (BME), is based on the end-to-end baseline model, E2E-Spot, as presented in [10]? citation.

We generate several variants of the E2E-Spot model to create a candidate model set and propose a strategy for selecting appropriate model members from this set while assigning appropriate weights to each selected model. More details can be found in [35]? citation.

The resulting ensemble model takes into account uncertainties in event length, optimal network architectures, and optimizers, making it more robust than the baseline model. Our approach has the potential to handle various video event analysis tasks. Explain: - Boosted Model Ensembling. - Candidate model generation. - Model selection and ensemble weighting. - The importance of model diversity. You don't need to describe every implementation detail. The important point is that it represents another successful approach to the same challenge. 3.3 Convolutional Temporal Modeling Explain the principles behind 3D CNNs.

Discuss: - 2D convolution extracts spatial features. - 3D convolution operates over spatial and temporal dimensions. - Temporal convolutions aggregate information from neighboring frames. - Slow fusion allows spatial feature extraction before temporal processing. Then explain the advantages of this approach, such as local temporal processing and efficient hierarchical feature extraction.

This subsection prepares the reader for your baseline architecture.

C. Transformer-Based Temporal Modeling

Introduce Transformers and self-attention.

Explain: - Temporal tokens. - Query, key and value representations. - Self-attention. - Positional embeddings. - Temporal aggregation. Discuss relevant applications of Transformers in video understanding and temporal action localization.

Finally, explain that your work investigates their use in ball action spotting, specifically as an alternative to the 3D CNN temporal encoder.

Important: Your actual Transformer architecture and its hyperparameters should not be described here. Those belong in Proposed Method.

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D. Transformer-Based Video Modeling

Introduce self-attention, positional information, and the use of Transformer encoders to model temporal sequences. Include work on temporal action spotting and video modeling that is technically relevant to your architecture. For instance, Transformer-based temporal reasoning has already been explored in soccer action spotting, so your paper should distinguish your particular replacement and evaluation from prior work rather than claim to introduce Transformers to the task.

E. C. Transformer-Based Temporal Modeling

To investigate the use of self-attention for temporal feature modeling, we replace the original 3D convolutional encoder with a Transformer encoder while retaining the shared 2D visual feature extractor. The input video is divided into groups of three consecutive grayscale frames, and each group is processed independently by the 2D encoder. The resulting feature maps are projected to a common feature dimension of 192 channels.

Unlike the original architecture, which performs temporal modeling on spatial feature maps, the proposed approach first applies adaptive average pooling to each feature map, producing a sequence of five 192-dimensional feature vectors for a 15-frame input. Learned positional embeddings are added to the sequence to provide information about the temporal ordering of the feature vectors.

The resulting sequence is processed by a Transformer encoder consisting of two layers with eight attention heads and an embedding dimension of 192. Self-attention allows each temporal token to incorporate information from the other tokens in the input sequence. The encoded representations are then aggregated by temporal average pooling to obtain a single 192-dimensional video representation.

Finally, a linear classification layer maps this representation to the output action classes. This architecture replaces the original 3D convolutional temporal encoder and its subsequent feature projection and concatenation with a Transformer-based temporal encoder and average-pooling operation. The effect of this architectural change is evaluated using the same action spotting task and evaluation metric as the baseline.

F. Convolutional Temporal Modeling

G. Transformer-Based Video Modeling

III. Proposed Method

This is probably the most important section of your paper.

The reader should understand exactly what you implemented and how the two architectures differ.

I recommend five subsections.

A. Task Definition

Mathematically define the problem.

Explain: - Input: a sequence of video frames. - Output: probabilities for the ball action classes. - Labels: actions associated with timestamps. - Prediction: whether an action occurs at the center timestamp of the input sequence. You should explicitly explain your label construction.

For example, if you use a 0.6-second positive labeling window, explain how frames within this interval are assigned positive labels.

You can introduce notation such as:

$$X = \{I_1, I_2, \ldots, I_T\}$$

where I_t represents a video frame.
The model predicts:

$$\hat{y} = f(X; \theta)$$

where $\hat{y}$ contains the predicted action probabilities. Define the input video sequence, the action classes, the timestamp-based labels, and the prediction target. Explain whether the model predicts the action at the center frame or predicts a temporal sequence of action probabilities.

Your draft describes labels expanded around the annotated timestamp. Make that explicit, including the window size, because it affects what the model is learning.

This novel task of ball action spotting consists in localizing the exact time when two types of actions related to the soccer ball occur: pass and drive. These actions are anchored with a single timestamp corresponding to the exact time the ball leaves a player's foot for a pass event and when a player touches the ball to control it for a drive event.
Ball action spotting involves identifying both the timing and type of ball-related actions across 12 classes in our dataset. Each action is marked by a single timestamp. In contrast to the SoccerNet Action Spotting Challenge, our videos feature a higher density of actions.

B. Soccer Action Spotting

ball action spotting, focusing on retrieving all timestamps related to the soccer ball change of state,

C. Ball Action Spotting

The SoccerNet Ball Action Spotting Challenge was introduced to address the problem of temporally localizing ball-related actions in broadcast soccer videos. Unlike general action spotting, ball-related events such as passes and drives are often densely distributed and require precise temporal localization.

Among the solutions proposed for the 2023 challenge, Baikulov achieved first place with an efficient deep learning architecture combining two-dimensional and three-dimensional convolutional operations [5]. The proposed architecture follows a slow-fusion strategy, in which a shared 2D convolutional encoder independently extracts visual features from groups of consecutive grayscale frames. These features are subsequently processed by a 3D convolutional encoder to capture temporal information. The resulting representations are aggregated through generalized mean pooling and used for multilabel action classification.

In addition to the model architecture, the solution introduced a multi-stage training strategy involving pre-training on the SoccerNet Action Spotting dataset, transfer learning, and fine-tuning with longer video sequences. The use of out-of-fold predictions for sample selection and an efficient video loading pipeline further contributed to the final performance. The solution achieved 86.47% mAP@1 on the challenge set [5].

Another notable approach was proposed by Wang et al., who obtained second place in the 2023 challenge with Boosted Model Ensembling (BME) [**?**]. Their method is based on the E2E-Spot baseline and focuses on improving performance through the generation and combination of multiple candidate models. These models differ in their training sample configurations, feature extraction architectures, and optimization strategies. The candidate architectures include RegNet and EfficientNet backbones incorporating Gate-Shift Modules, while the training strategies consider AdamW and stochastic weight averaging.

Rather than relying on a conventional fixed ensemble, BME iteratively selects candidate models and determines their contribution according to improvements in validation mAP@1. The resulting ensemble achieved 86.37% mAP@1 on the challenge test set [**?**].

These approaches demonstrate the importance of both spatiotemporal feature extraction and training strategies for accurately spotting ball-related events. In particular, the slow-fusion architecture proposed by Baikulov provides the convolutional baseline considered in this work. We investigate an alternative approach to temporal feature modeling by replacing its 3D convolutional encoder with a Transformer encoder, while retaining the shared 2D visual feature extractor.

D. Baseline Architecture

Describe Baikulov's architecture as the reference architecture you reproduce or adapt.

Include a diagram.

Explain the processing pipeline: 1. Sample frames from the video. 2. Group neighboring grayscale frames into stacks of three. 3. Apply the shared 2D EfficientNetV2-B0 encoder. 4. Process the resulting feature maps using the 3D temporal encoder. 5. Aggregate features using pooling. 6. Predict action probabilities using a classification layer. Include tensor dimensions wherever possible.

For example, assuming five stacks and a 192-channel projected representation, describe the actual dimensions produced by your implementation rather than relying on this example.

E. Transformer-Based Temporal Encoder

This is your main architectural contribution.

Explain the architecture step by step.

Based on your previous draft: 1. Divide the 15-frame input into five groups of three frames. 2. Process each group using the shared 2D encoder. 3. Project the extracted features to 192 dimensions. 4. Apply adaptive average pooling to obtain five temporal feature vectors. 5. Add learned positional embeddings. 6. Process the sequence with a Transformer encoder. 7. Apply temporal average pooling. 8. Use a linear classifier to predict the action classes. Include the actual Transformer configuration: Parameter Value Embedding dimension 192 Number of layers 2 Attention heads 8 Number of temporal

tokens 5 Positional encoding Learned Temporal aggregation Average pooling

Also report the feed-forward dimension, dropout, normalization, and activation function from your implementation.

Explain why you chose these values, if you have experimental justification.

F. Architectural Differences

I strongly recommend adding this subsection.

A table will make your contribution much easier to understand. Component CNN baseline Transformer 2D backbone EfficientNetV2-B0 Same Input frames 15 15 Frame grouping 3 3 Temporal encoder 3D CNN Transformer Spatial aggregation As implemented in baseline Adaptive average pooling Temporal aggregation Baseline pooling Average pooling Classifier Linear Linear

Verify every entry against your actual code.

In particular, explain that spatial pooling before attention changes the feature representation as well as the temporal modeling operation. This distinction is important for interpreting the comparison.

G. Training and Inference

Describe the training and prediction pipelines.

Include: - Input resolution. - Frame sampling rate. - Number of input frames. - Label construction. - Loss function. - Optimizer. - Learning rate and scheduler. - Batch size. - Data augmentation. - Initialization and pretrained weights. - Number of epochs. - Prediction procedure. - Postprocessing. For inference, explain how frame-wise probabilities are converted into timestamp predictions.

For example, describe Gaussian smoothing and peak detection if you retain those operations.

H. Baseline Architecture

Research question is supposed to be about replacing the temporal 3D CNN with a Transformer Describe the existing pipeline with a diagram and a concise explanation: 1. Sample frames from the video. 2. Stack neighboring grayscale frames into groups of three. 3. Extract visual features using the shared 2D encoder. 4. Process the feature sequence using the 3D temporal encoder. 5. Pool the features and predict action probabilities. The draft already contains most of this information. Add tensor dimensions at each major stage, where possible, so that the reader can reproduce the architecture.

I. Baseline Architecture

This should be the main contribution. Specify: - Which features enter the Transformer: raw frame embeddings, spatially pooled 2D features, or intermediate CNN feature maps. - The projection from visual feature dimension to Transformer embedding dimension. - The positional encoding method. - Number of Transformer layers and attention heads. - Feed-forward dimension, normalization, dropout, and activation. - How the temporal representation is pooled or selected for classification. Explain which parts remain unchanged from the baseline and which are replaced.

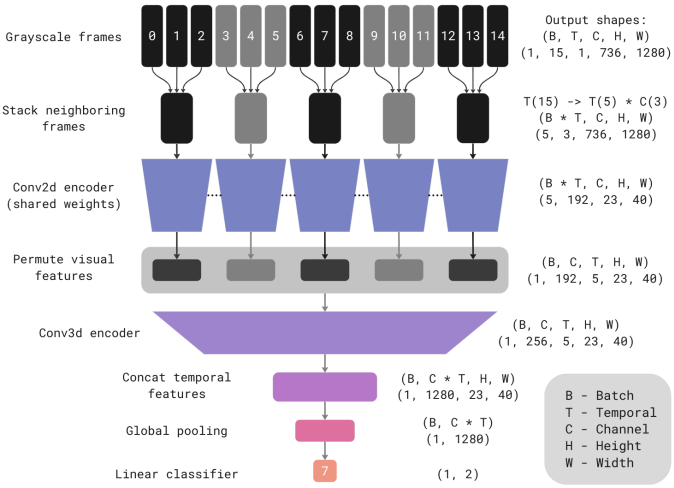


Fig. 1. Model second solution

J. Training and Inference

Describe the common training setup, loss, label expansion, augmentation, and prediction postprocessing. If the Transformer requires different optimization settings or a different sequence length, report those differences clearly.

A key methodological choice is whether you train the Transformer from scratch or initialize it from pretrained weights. That choice affects how you interpret the results, particularly with limited labeled data.

K. Temporal Feature Representation

L. Transformer Encoder

M. Prediction and Postprocessing

IV. Experimental Setup

The purpose of this section is reproducibility.

Another researcher should be able to understand how you performed the experiments.

A. Dataset

Describe the actual challenge dataset you used.

Include: - Dataset source. - Number of matches. - Number of annotated events. - Number of action classes. - Video resolution and frame rate. - Training, validation and test splits. - Challenge evaluation set. Be especially careful about the challenge year: your previous draft mixed 2023 and 2024 dataset descriptions.

Use the correct dataset statistics for your experiment. The dataset is provided by Footovision and is composed of 7 games with a total of 11,041 annotated timestamps, as well as 2 extra segregated games for the challenge set.

The dataset is [6]

The data consists of 7 videos of English Football League games, each broadcast in 720p quality. We don't provide extracted features to encourage the development of end-to-end methods. The challenge set is composed of 2 separate games. Additionally, participants are encouraged to leverage the 500

unannotated games from the Action Spotting challenge as a resource for this task.

The dataset available is relatively small compared to the action spotting task, encouraging participants to explore semi-supervised, selfsupervised, or transfer learning methods using the 500 unlabeled broadcast games.

A significant enhancement this year is the inclusion of 12 classes of ball actions, requiring more precise classification. Consequently, simple categorizations like "pass" are insufficient as specific pass types must be clearly identified.

We provide annotation for all two types of soccer ball actions: Pass, Drive. In 2024, we extend it to 12 different classes of soccer ball actions: Pass, Drive, Header, High Pass, Out, Cross, Throw In, Shot, Ball Player Block, Player Successful Tackle, Free Kick, Goal

B. Implementation Details

Describe your technical environment.

For example: - PyTorch version. - GPU hardware. - CUDA version. - Input resolution. - Batch size. - Optimizer. - Training duration. - Mixed precision. - Random seeds. Also report the relevant model hyperparameters.

C. Evaluation Metrics

Explain mAP@1.

Describe: - Average Precision. - Mean Average Precision across action classes. - The one-second temporal tolerance. - The official evaluation protocol. If you measure inference speed or memory, explain those metrics here too. Unlike the SoccerNet Action Spotting Challenge, that uses the tight average-mAP ([1-5] seconds tolerance), this challenge focuses on accurately spotting the ball events with the mAP@1 (1 second tolerance).

D. Experimental Configurations

Clearly identify your experiments.

I suggest the following table: Experiment Temporal encoder Purpose E1 3D CNN Baseline E2 Transformer Proposed architecture E3 Transformer, alternative configuration Ablation, if available

Try to keep the experiments controlled.

Ideally, the baseline and Transformer should use the same: - Dataset split. - Input frames. - 2D backbone. - Training data. - Label construction. - Evaluation procedure. - Postprocessing. If you cannot keep everything identical, document the differences.

The goal here is to make the experiment reproducible and the comparison defensible. Subsection What it should contain Dataset Number of matches, classes, annotations, and data split Implementation Framework, hardware, input resolution, model settings Evaluation metrics mAP@1 and any efficiency metrics Experimental configurations Baseline, Transformer variant, and ablations Training protocol Initialization, optimizer, schedule, seeds, and validation procedure

Keep the primary comparison as controlled as possible: use the same data splits, input frames, 2D feature extractor, label construction, postprocessing, and evaluation implementation for both models.

If you change several things at once — for example, the temporal encoder, the input length, and the loss — then a performance difference cannot be attributed confidently to the Transformer.

E. Implementation Details

F. Evaluation Metrics

G. Experimental Configurations

V. Results and Discussion

This section should answer your research question, not just present numbers.

I recommend five subsections.

A. Main Comparison

This isn't true, TODO: rewrite and check

We use the 3D CNN-based architecture from the 2023 challenge as the reference for our design. Since the proposed model was trained on the 2024 data, results from the two challenge years are not a controlled comparison of temporal architectures. Differences in dataset composition, training data, and evaluation conditions must be considered when interpreting the results.

This is your most important results table.

For example: Architecture mAP@1 Parameters Inference time 3D CNN baseline TBD TBD TBD Transformer TBD TBD TBD

Use actual measurements, and clarify whether inference time is measured per frame, sequence, or video.

Discuss: - Which architecture performs better? - What is the absolute difference in mAP? - Are the differences consistent across classes or folds? - Does the Transformer introduce computational overhead? Don't interpret a small difference as significant without suitable repeated experiments.

B. Effect of Temporal Context

Investigate how the length of the input sequence affects performance.

For example: - 15 frames. - 33 frames. - Other sequence lengths, if you tested them. Explain whether longer temporal context improves spotting performance.

If you compare sequence lengths, make sure the sampling rate and temporal coverage are clearly specified.

C. Ablation Study

Investigate individual architectural choices.

Possible experiments: - Transformer depth. - Number of attention heads. - Embedding dimension. - Positional embeddings. - Temporal pooling strategy. You don't need to test all of these. Select the experiments you have actually performed. 6.4 Error Analysis This section is often overlooked but can make your paper considerably stronger.

Analyze: - False positives. - Missed events. - Closely spaced actions. - Confusion between action classes. - Temporal localization errors. Include qualitative examples if possible.

For instance, show a short video timeline with ground-truth timestamps and predicted timestamps.

D. Limitations

Be honest about the limitations.

Potential points: - Limited size of the annotated dataset. - Small number of temporal tokens. - Spatial information lost through global pooling before attention. - Computational requirements. - Lack of extensive hyperparameter tuning. - Differences between the original winning solution and your reproduction. Only include limitations that genuinely apply. Your current proposed subsections are sensible, but each should answer a specific question.

E. Main Comparison

Compare the CNN and Transformer using the same evaluation protocol. Report mAP@1, absolute difference in percentage points, and, if feasible, variation across random seeds or folds.

F. Effect of Temporal Context

Test whether performance changes when the input sequence becomes longer. To isolate the effect of the encoder, use comparable temporal windows for both architectures.

G. Ablation Study

Test one design choice at a time, such as embedding dimension, number of Transformer layers, or positional encoding. Choose only a few ablations that help explain the main result.

H. Error Analysis

Investigate false positives, missed actions, closely spaced events, and differences between action classes. Use examples or per-class metrics where possible.

I'd also report inference latency and peak GPU memory if efficiency is part of the research question. Parameter count alone is not a sufficient measure of computational efficiency.

One distinction matters for your current draft: the leaderboard table is useful context, but it is not a substitute for a controlled baseline comparison. Your model must be evaluated on the same split and with the same protocol before you can attribute a change to the temporal encoder.

We evaluate the proposed Transformer-based model on the 2024 ball-action spotting challenge using the official evaluation protocol. We report the primary challenge metric and describe the evaluation split and inference procedure.

Here we need to compare to other results.

After the table

I. Effect of Temporal Context

J. Ablation Study

K. Error Analysis

We examine the model's predictions to identify common false positives, missed actions, and temporal localization errors. Where ground-truth labels are available, we also analyze performance by action class.

TABLE I

BALL ACTION SPOTTING LEADERBOARD. THE MAIN METRIC FOR THE LEADERBOARD AND BEST PERFORMANCES ARE IN BOLD. TEAM NAMES WITH A SUPERSCRIPT HAVE PROVIDED A SUMMARY THAT MAY BE FOUND IN APPENDIX **??** OR IN SECTION **??** FOR THE WINNING TEAM.

Participants	mAP @1	mAP @2	mAP @3	mAP @4	mAP @5	Average-mAP tight
Ruslan Baikulov^{B1}	**86.47**	**87.98**	**88.28**	**88.18**	**87.95**	**87.91**
FDL@ZLab^{B2}	83.39	85.19	85.81	86.00	86.19	85.45
BASIK^{B3}	82.06	83.39	83.86	84.04	83.91	83.57
FC Pixel Nets^{B4}	81.89	83.22	83.97	83.85	84.02	83.50
play	79.74	82.58	84.06	84.49	84.34	83.29
Baseline [**?**]*	62.72	69.24	72.57	74.29	74.80	71.21

TABLE II

BALL ACTION SPOTTING LEADERBOARD. THE MAIN METRIC FOR THE LEADERBOARD AND BEST PERFORMANCES ARE IN BOLD.

L. Limitation

The current comparison does not isolate the effect of replacing 3D convolutions with self-attention. The proposed model also changes spatial and temporal feature aggregation. A controlled comparison using both architectures trained and evaluated under the same conditions would be needed to attribute performance differences more specifically to the temporal modeling approach

VI. CONCLUSION AND FUTURE WORK

A. Conclusion and Future Work

Keep this section relatively short.

Structure it in three paragraphs.

Paragraph 1 — Objective and method

Summarize the problem and your proposed architecture.

Paragraph 2 — Main findings

Report the actual experimental results and explain what they demonstrate about CNN-based and Transformer-based temporal modeling.

Paragraph 3 — Future work

Possible directions: - Spatial-temporal attention instead of spatially pooled temporal tokens. - More efficient Transformer architectures. - Longer temporal sequences. - Pretraining on larger soccer video datasets. - Combining convolutional and attention-based temporal representations. Avoid introducing new experimental claims in the conclusion.

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